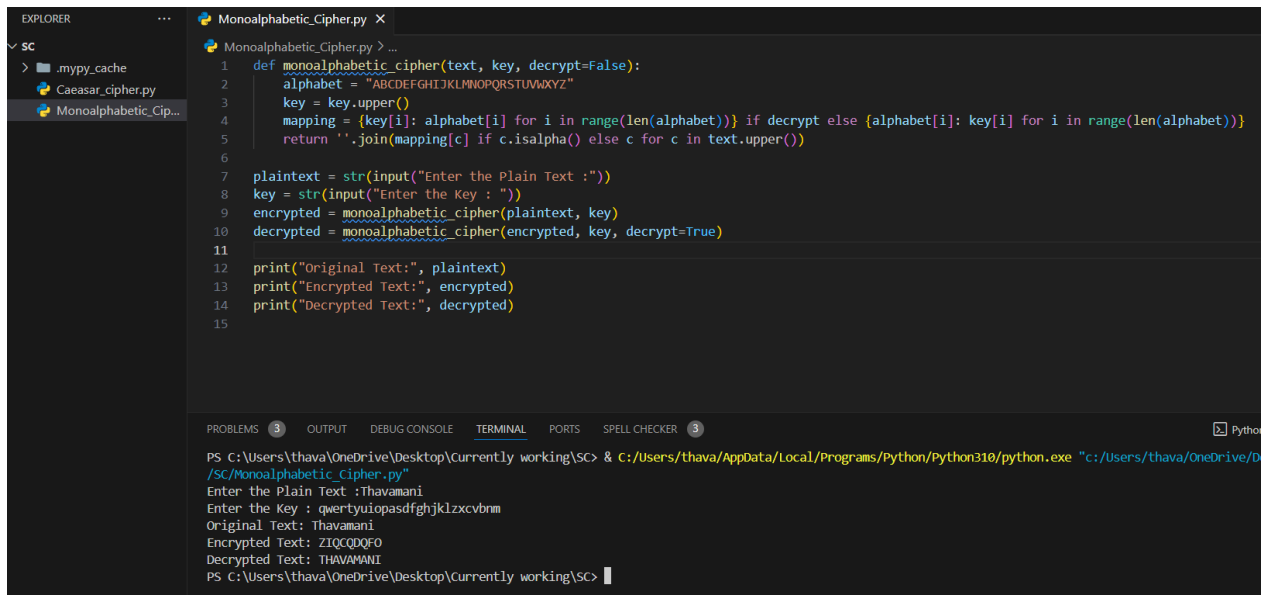


Exercise 2 – Monoalphabetic Cipher

Code :

```
def monoalphabetic_cipher(text, key, decrypt=False):  
    alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"  
    key = key.upper()  
    mapping = {key[i]: alphabet[i] for i in range(len(alphabet))} if decrypt else  
    {alphabet[i]: key[i] for i in range(len(alphabet))}  
    return ''.join(mapping[c] if c.isalpha() else c for c in text.upper())  
  
plaintext = str(input("Enter the Plain Text :"))  
key = str(input("Enter the Key : "))  
encrypted = monoalphabetic_cipher(plaintext, key)  
decrypted = monoalphabetic_cipher(encrypted, key, decrypt=True)  
  
print("Original Text:", plaintext)  
print("Encrypted Text:", encrypted)  
print("Decrypted Text:", decrypted)
```

Output:



The screenshot shows a code editor with a file named 'Monoalphabetic_Cipher.py'. The code is as follows:

```
1 def monoalphabetic_cipher(text, key, decrypt=False):  
2     alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"  
3     key = key.upper()  
4     mapping = {key[i]: alphabet[i] for i in range(len(alphabet))} if decrypt else {alphabet[i]: key[i] for i in range(len(alphabet))}  
5     return ''.join(mapping[c] if c.isalpha() else c for c in text.upper())  
6  
7     plaintext = str(input("Enter the Plain Text :"))  
8     key = str(input("Enter the Key : "))  
9     encrypted = monoalphabetic_cipher(plaintext, key)  
10    decrypted = monoalphabetic_cipher(encrypted, key, decrypt=True)  
11  
12    print("Original Text:", plaintext)  
13    print("Encrypted Text:", encrypted)  
14    print("Decrypted Text:", decrypted)  
15
```

The terminal output shows the execution of the program:

```
PS C:\Users\thava\OneDrive\Desktop\Currently working\SC> & C:/Users/thava/AppData/Local/Programs/Python/Python310/python.exe "c:/Users/thava/OneDrive/D  
/SC/Monoalphabetic_Cipher.py"  
Enter the Plain Text :Thavamani  
Enter the Key : qertyuiopasdfghjklzxcvbnm  
Original Text: Thavamani  
Encrypted Text: ZIQCQDQFO  
Decrypted Text: THAVAMANI  
PS C:\Users\thava\OneDrive\Desktop\Currently working\SC>
```